## Interpreting LLM–Brain Alignment Through Shared Inductive Biases

**Introduction.** Large Language Models (LLMs) allow us to probe human language processing in a direct way by comparing their sentence representations to activity in the human Language Network (LN). In this setting, high "brain scores" – correlations between LLM sentence representations and LN fMRI responses – are taken as evidence of shared computational principles (Goldstein et al., 2022). However, such scores can be inflated by trivial properties such as sentence length (Feghhi et al., 2024), and mechanistic differences between biological and artificial networks make it unclear which aspects of sentence processing this alignment reflects. We therefore ask how to embed brain scores in a principled framework where they provide evidence about *shared inductive biases* (IB) between LLMs and the LN. Inductive biases are a natural level of analysis for alignment work, but have so far played only an informal role. Our contribution is twofold. First, motivated by the goal of using high alignment as evidence about key computational components of human language processing, we propose a framework for studying shared IBs in language tasks. Second, we design experiments that begin to validate this framework and demonstrate how it can be applied to analyze alignment results. We focus on the Pereira et al. (2018) fMRI dataset ($N = 384$ short read English sentences grouped by topic); we keep only the top-512 most LN-aligned LLM units, as in AlKhamissi et al. (2024), and fit voxel-wise ridge regression decoders from LLM to fMRI.

**Bias Formalization.** We want a notion of IB that lets us compare biological and artificial models at the level of sentence processing rather than mechanisms. We formalize a language task as $T = (\mathcal{X}, \mathcal{Y}, D_{\mathcal{X}\times\mathcal{Y}}, \ell)$, where $\mathcal{X}$ is the space of inputs (sentences) and $\ell$ the loss. Let $\mathcal{H} \subseteq \{f : \mathcal{X} \to \mathcal{Y}\}$ be a hypothesis space, and define the solution set $\mathcal{S}_T(\varepsilon) = \{f \in \mathcal{H} : R_T(f) \leq \varepsilon\}$ with risk $R_T(f) = \mathbb{E}_{(x,y)\sim D}[\ell(f(x), y)]$. A learner $\lambda = (\mathcal{A}, \mathcal{O}, \Phi, S_{\text{train}})$ — architecture, optimizer, objective, and training data — selects a function $f_\lambda \in \mathcal{H}$. We define its *inductive bias* as the subset $\Omega \subseteq \mathcal{H}$ of solutions in $\mathcal{S}_T(\varepsilon)$ that $\lambda$ returns with high probability across draws of $S_{\text{train}}$. Thus, $\Omega$ is determined by the interaction of all components of $\lambda$, not only the architecture, accounting for alignment with untrained and trained models (AlKhamissi et al., 2025). We further distinguish *first-order* biases $\Omega^{(1)}$, which specify the support of solutions (e.g., $\lambda_0$ must encode position in the sequence, so any $f \in \mathcal{S}_T(\varepsilon)$ respects positional constraints), from *second-order* biases $\Omega^{(2)} \subseteq \Omega^{(1)}$, which describe preferred geometries among those solutions (e.g., $\lambda_0$ encodes position in an absolute rather than relative code). Raw brain scores $r(x)$ mix biases of different kinds, so we decompose $r(x) = r_\parallel(x) + r_\perp(x)$, where $r_\parallel$ is the projection onto the subspace spanned by simple objective features $\phi_0$ (Table 1) and $r_\perp$ is the residual that reflects the learner's inductive preferences.

**Empirical Validation.** We validate the hierarchical distinction between $\Omega^{(1)}$ and $\Omega^{(2)}$ constraints using temporal integration: while context sensitivity is an $\Omega^{(1)}$ requirement for any language model (AlKhamissi et al., 2024), we manipulate the attention mechanism of an untrained GPT-2 with an exponential decay factor $\alpha$ and find that alignment in $r_\perp$ peaks under long-horizon integration ($\alpha < 0$), significantly outperforming strongly local models (Figure 1, $p < 0.001$), suggesting *long-range maintenance* of context as a second-order bias $\Omega^{(2)}$ of LN. We then apply the framework to prediction error (PE) theories by testing whether LN responses track absolute probabilistic PE or a signal relative to discourse-level expectations. We encode these alternatives as IBs by defining sentence-level surprisal—raw contextual surprisal versus static and moving variants that approximate discourse-adapted expectations. Absolute surprisal yields the strongest alignment with LN activity (mean $r \approx .12$, max $r \approx .48$, all $p < 0.05$), whereas normalization reliably weakens correlations. Finally, we show how the separation of $r_\perp$ from $r_\parallel$ results in non-trivially different layer-wise alignment (Figure 2). Taken together, these results show how brain scores can be interpreted as evidence about inductive biases, and how targeted training manipulations can probe them.

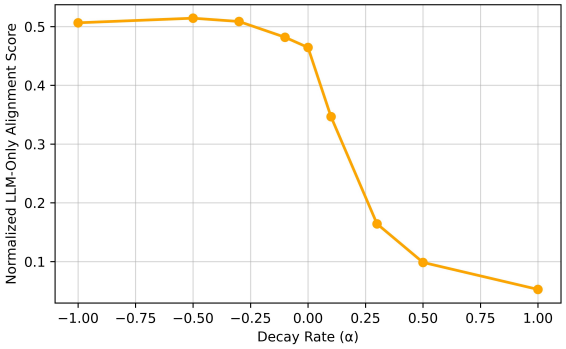


Figure 1: Effect of Decay Rate on Normalized Alignment (GPT2, Untrained). Negative $\alpha$ values, which preserve more strongly long-range context, yield similarly high scores, whereas even modestly positive decay (strong locality) leads to a sharp drop in alignment. This steep degradation suggests that LN-like responses are better captured by architectures that maintain broad, long-horizon sentence context than by ones that heavily privilege nearby words.

| Feature | Description |
|---|---|
| Sentence Length (SL) | Number of words in sentence. |
| Sentence Position (SP) | One-hot position vector. |
| Word Position (WP) | Average word position (9-dim). |
| Mean Log Freq. | Average Zipf frequency. |
| Std Log Freq. | Std of Zipf frequency. |
| Type-Token Ratio | Unique words / Total words. |

Table 1: Objective Features. Following Feghhi et al. (2024), we use shallow, stimulus-defined features as the baseline for the $r_{\|}(x)$ score. A feature is "objective" if it is a property of the sentence itself and can be computed without modeling any substantive neural processing. For example, a positional bias may be implemented in many ways in a model, but the bias itself can already be captured by a simple counter.

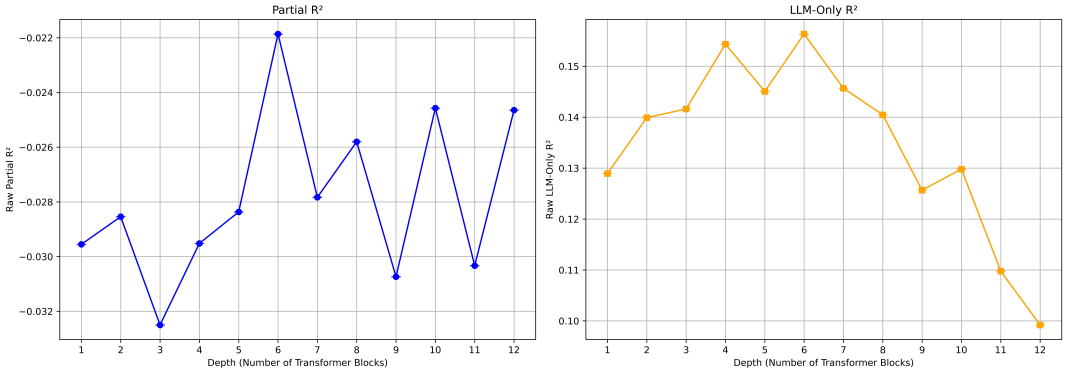


Figure 2: Layer wise alignment for trained GPT-2 on Pereira2018 with topic wise cross validation (train test split by topic, which substantially lowers absolute scores). Both panels show raw explained variance $R^2$ (not correlations, not ceiling normalized). Right: LLM only features yield the well-known, broad mid layer peak (Antonello et al., 2023). It is thought that representations are progressively built up through early layers, become especially rich around the middle where alignment peaks, and then are increasingly compressed in later layers to support next token prediction. Left: after regressing out objective features, partial $R^2$ on $r_{\perp}$ is small and often slightly negative, reflecting the sensitivity of test $R^2$ to strong ridge regularization ($\alpha = 10^3$) on the top 512 localized units. The curve is sharper and noisier, with only a modest bump at layers 6–8, suggesting that once baselines are removed, shared inductive bias reduces to a narrow mid hierarchy compression effect rather than a broad intermediate layer property.

# References


AlKhamissi, B., Tuckute, G., Bosselut, A., & Schrimpf, M. (2024). Brain-like language processing via a shallow untrained multihead attention network. *arXiv preprint arXiv:2406.15109*. https://doi.org/10.48550/arXiv.2406.15109

AlKhamissi, B., Tuckute, G., Tang, Y., Binhuraib, T. O. A., Bosselut, A., & Schrimpf, M. (2025, November). From language to cognition: How LLMs outgrow the human language network. In C. Christodoulopoulos, T. Chakraborty, C. Rose, & V. Peng (Eds.), *Proceedings of the 2025 conference on empirical methods in natural language processing* (pp. 24332–24350). Association for Computational Linguistics. https://doi.org/10.18653/v1/2025.emnlp-main.1237

Antonello, R., Vaidya, A., & Huth, A. G. (2023). Scaling laws for language encoding models in fmri. *Advances in Neural Information Processing Systems*, *36*.

Feghhi, E., Hadidi, N., Song, B., Blank, I. A., & Kao, J. C. (2024). What are large language models mapping to in the brain? a case against over-reliance on brain scores. https://doi.org/10.48550/arXiv.2406.01538

Goldstein, A., Zada, Z., Buchnik, E., Schain, M., Price, A. R., Aubrey, B., Nastase, S. A., Feder, A., Emanuel, D., Cohen, O., Moshe, H., Noy, N., Hasson, U., & Honey, C. J. (2022). Shared computational principles for language processing in humans and deep language models. *Nature Neuroscience*, *25*(3), 369–380. https://doi.org/10.1038/s41593-022-01026-4

Pereira, F., Lou, B., Pritchett, B., Ritter, S., Gershman, S. J., Kanwisher, N., Botvinick, M., & Fedorenko, E. (2018). Toward a universal decoder of linguistic meaning from brain activation. *Nature Communications*, *9*(1), 963. https://doi.org/10.1038/s41467-018-03068-4